

INSTAGRAM USAGE & LIFESTYLE

A Social Media Data Analysis Project

DEPI Round 4 — Graduation Project

Exploring the relationship between social media usage,
lifestyle habits, and digital productivity

Python (ETL) • SQL Server (Relational Modeling) • Power BI (Dashboarding)

1. Project Overview

Social media plays a major role in daily life, but excessive usage may negatively affect productivity and time management. This project analyzes social media usage patterns and their relationship with lifestyle and wellbeing indicators — sleep, stress, happiness, physical activity, and diet — to identify behavioral trends and possible indicators of digital habit formation and dependency.

The analysis focuses on how users interact with the Instagram platform, how much time they spend across its surfaces (Feed, Reels, Explore, Stories, Messages), and how this behavior relates to demographic, health, and monetization factors. The final outcome is an interactive Power BI dashboard that presents these insights through six linked report pages, supported by a Python-based data cleaning pipeline and a normalized SQL Server data model.

Dataset Source

Kaggle — "Social Media User Analysis" (kaggle.com/datasets/rockyt07/social-media-user-analysis), filtered to the Instagram usage & lifestyle subset used for this project.

Team Members

Nour Ahmed Youssef	Team Leader
Abdelrahman Mohamed Hamza	Member
Mahmoud Ahmed Abbas	Member
Mohamed Amr Mohamed	Member
Mohamed Nagah Mohamed	Member
Youssef Emad Mohamady	Member

2. Project Plan & Methodology

2.1 Research & Analytical Questions

- How much time do users spend on Instagram daily, and on which surfaces (Feed, Reels, Explore, Stories, Messages)?
- Which demographic and lifestyle segments show the highest engagement and ad interaction?
- Does increased social media usage relate to sleep, stress, happiness, or physical activity levels?
- What patterns exist between income, subscription tier, and monetization behavior (ads viewed/clicked, premium adoption)?

2.2 Data Preparation (Python)

The raw dataset was downloaded from Kaggle via the Kaggle API inside a Google Colab notebook (`instagram_usage_lifestyle.py`) and loaded into pandas for cleaning. The pipeline performed the following steps:

- Removed redundant columns (e.g. a single-value `app_name` field).
- Engineered a new `age_group` feature, bucketing users into Gen Alpha, Gen Z, Millennial, and Gen X cohorts based on age.
- Converted binary Yes/No fields (`has_children`, `uses_premium_features`, `two_factor_auth_enabled`, `biometric_login_used`) into 1/0 flags for analysis.
- Normalized 14 categorical attributes — gender, country, age group, urban/rural, income level, employment status, education level, relationship status, diet quality, smoking, alcohol frequency, content type preference, preferred content theme, privacy setting level, and subscription status — into separate dimension tables, each with a surrogate ID key, replacing the raw text in the main table.
- Reordered and exported the final cleaned dataset as a single fact table (`fact_usage.csv`) alongside 15 dimension CSVs (e.g. `dim_gender`, `dim_country`, `dim_age_group`, etc.).

This star-schema-style cleaning (one fact table + multiple dimension tables) keeps the dataset compact, removes redundant text repetition, and prepares it for relational loading.

2.3 Database & Relational Modeling (SQL Server)

The cleaned fact and dimension tables were loaded into SQL Server, where primary and foreign key relationships were established between `fact_usage` and each dimension table (gender, country, age group, income level, employment status, education level, relationship status, diet quality, smoking, alcohol frequency, content type preference, content theme, privacy setting level, and subscription status). This relational model was then used as the live data source for the Power BI report, and for writing SQL queries to validate aggregates, segment users, and cross-check the dashboard's calculated metrics.

2.4 Dashboard Development (Power BI)

Insights were visualized in an interactive, multi-page Power BI dashboard with synchronized slicers for Age Group, Gender, and Country, allowing every page to be filtered consistently across the report. The dashboard is organized into six pages, each covering a distinct analytical theme, detailed in Section 3.

3. Power BI Dashboard — Page by Page

The dashboard contains 1.55M analyzed users and uses a consistent global filter panel (Age Group, Gender, Country) across all six pages. The pages below reflect the latest refresh of the report.

3.1 Executive Summary

Provides the top-line view of the platform: 1.55M total users, 232K premium users (15%) and 155K business accounts (10%). The usage trend chart shows the same monthly active-time fluctuations through 2025, and the geographic distribution still highlights the United States, India, and Brazil as the largest user bases. Time spent by content is now broken out as total minutes rather than an average, split almost identically between Explore, Reels, and Feed for both male and female users (28M / 42M / 70M respectively), while non-binary and 'prefer not to say' users contribute far smaller totals. The average engagement score by subscription tier is also more clearly differentiated than before, with Business and Premium accounts now averaging visibly higher engagement than Free accounts.

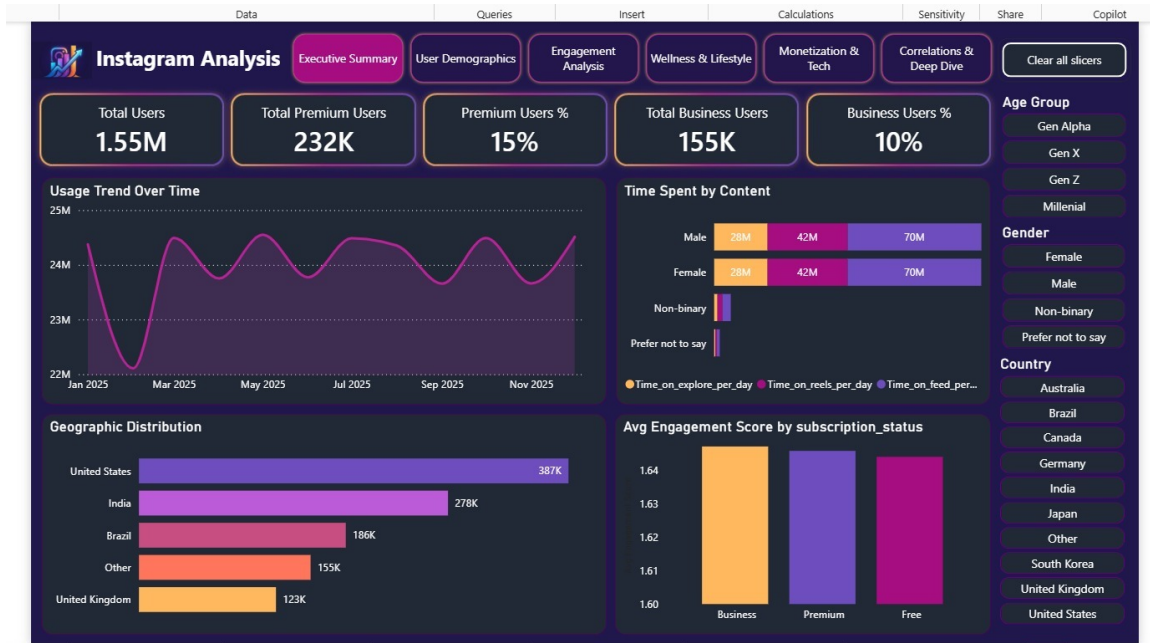


Figure 1 — Executive Summary page

3.2 User Demographics

Profiles the user base: average age of 39, Urban as the dominant living area, Male as the dominant gender, Full-time employment as the most common job status, and Single as the leading relationship status. Age segmentation is unchanged, with Gen X as the largest cohort (642K users), followed by Gen Z (439K) and Millennials (409K), while Gen Alpha remains a small fraction (58K). The employment breakdown now shows Full-time employed users (695K) and Students (340K) as the two largest groups, and the income tier and education-by-gender charts round out the demographic profile.



Figure 2 — User Demographics page

3.3 Engagement Analysis

Centers on platform behavior: users log 291M total daily minutes on Instagram platform-wide, send 59 weekly messages on average, hold an average of 2K followers, and view 101 stories on average, with a 24% ads click-through rate. Engagement is now shown in aggregate rather than per-user terms, and the ranking has flipped from earlier readings: Gen X generates the largest total volume of weekly DMs and likes (18M received, 17M sent, 10M average likes), ahead of Gen Z, Millennials, and Gen Alpha — consistent with Gen X being the largest cohort overall. Engagement depth stays nearly uniform across content types and subscription tiers, and user interest is spread almost evenly across the eight content themes (roughly 193K–195K users each), with Fitness and Tech still edging out the rest.

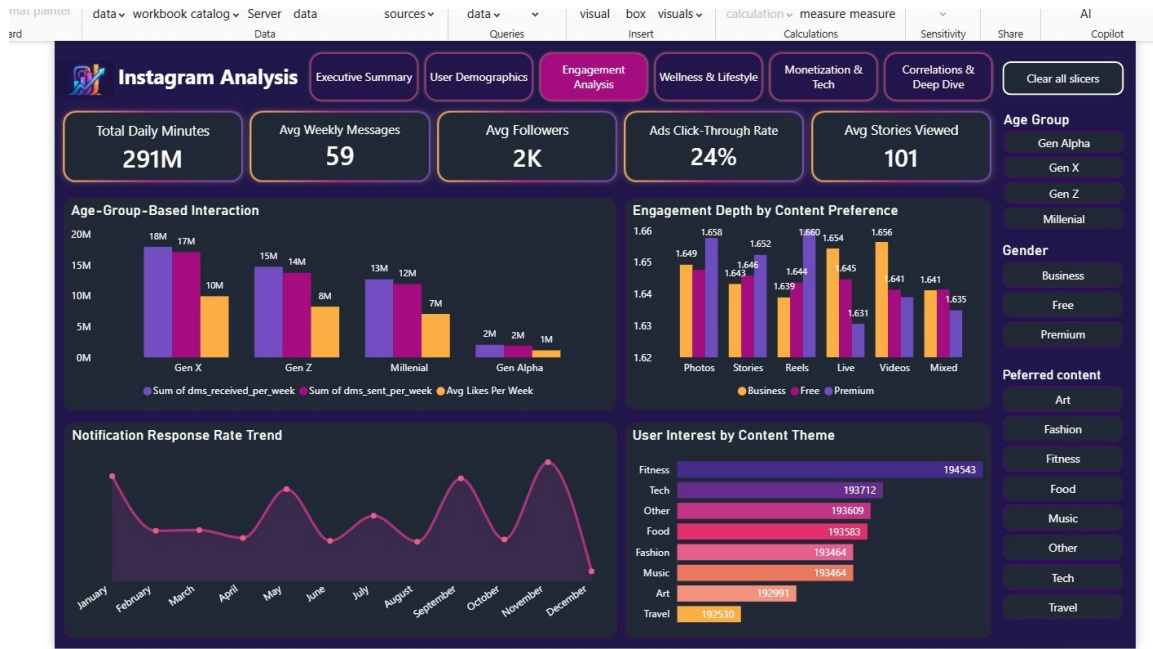


Figure 3 — Engagement Analysis page

3.4 Wellness & Lifestyle

Connects usage patterns to wellbeing indicators: average 8K daily steps, 7 hours of sleep, a stress score of 20, 10.33 sessions per user, and a happiness score of 5.5. The standout new finding is a clear inverse relationship between daily active minutes and self-reported happiness — users active for under 100 minutes a day report happiness scores as high as 8–9, while those active 500+ minutes a day cluster closer to 1–2, with a steady decline across the full range in between. Stress stays essentially flat (~7.0) across generations, and the user base is overwhelmingly non-smoking (1.16M 'No' vs 0.23M 'Yes' and 0.15M 'Former').

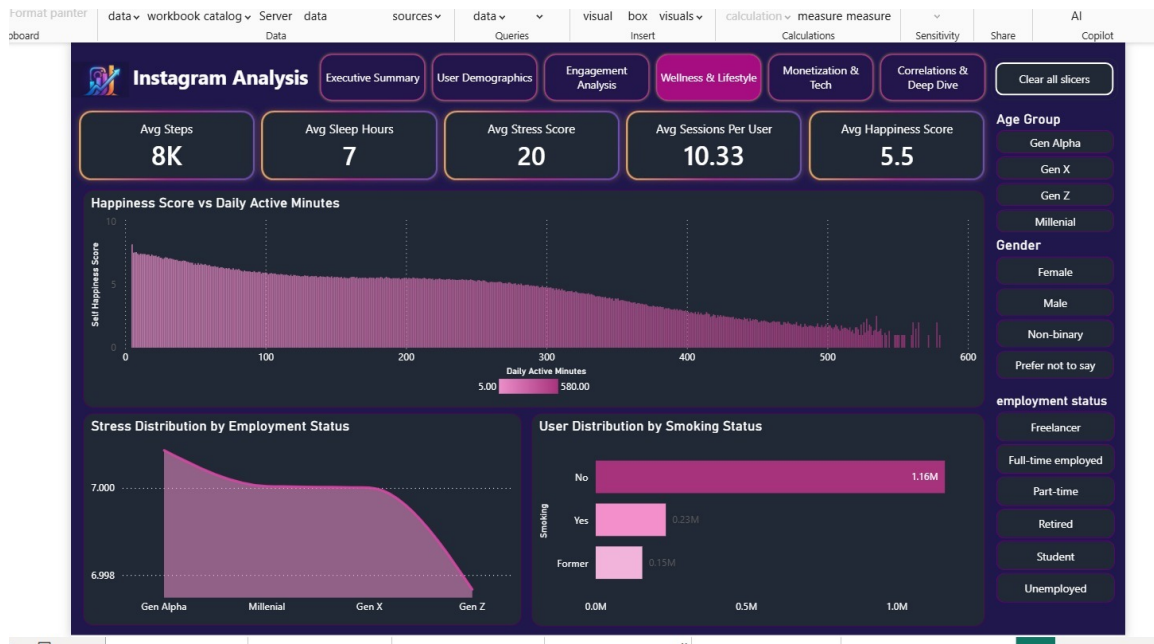


Figure 4 — Wellness & Lifestyle page

3.5 Monetization & Tech

Focuses on revenue and security signals: 31M total ads viewed, an average of 4.77 daily ads clicked, 620K users with biometric security adoption, 1M accounts with Two-Factor Authentication enabled, and 15% of users on a premium plan. Ad exposure still rises with age (Gen X views the most ads at 11.7M, Gen Alpha the least at 1.3M), and the subscriber base remains overwhelmingly Free-tier (74.95%). Total ads clicked by income level is now labeled with exact figures and reveals an inverse pattern: Middle (2.21M) and Lower-middle (1.84M) income users click the most ads in aggregate, while High-income users click the fewest (0.74M).

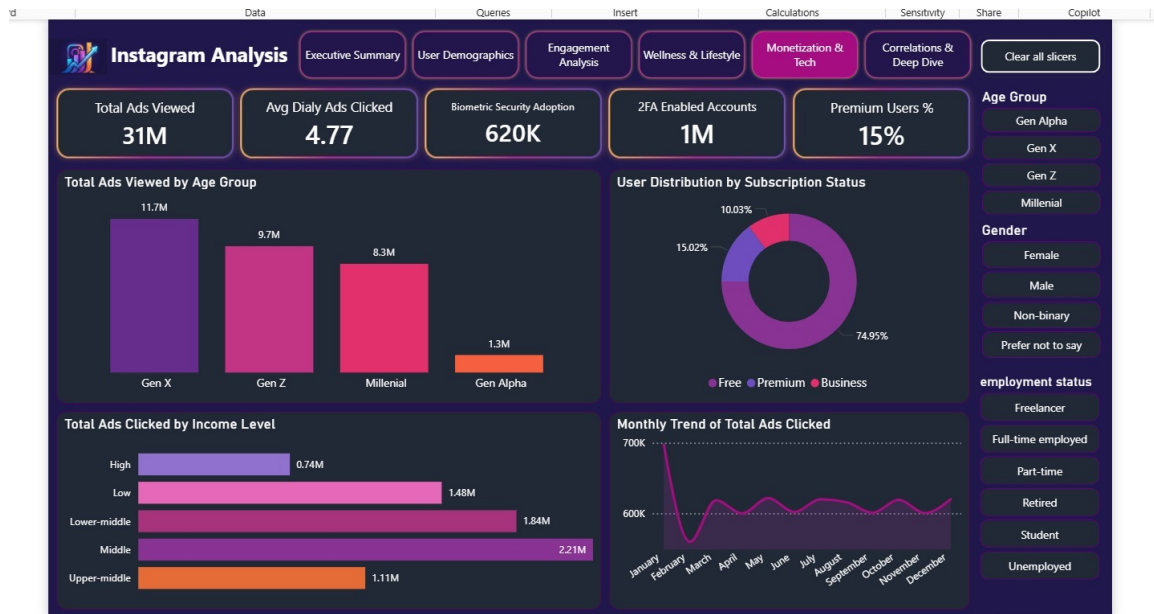


Figure 5 — Monetization & Tech page

3.6 Correlations & Deep Dive

An exploratory analysis page combining a decomposition tree (Total Users broken down by Country and Age Group), a scatter plot relating ad consumption to engagement across generations, and a Key Influencers visual identifying drivers of user_engagement_score. The KPI cards now surface the underlying averages directly — a 1.64 average engagement score, 188.23 average daily minutes, and a stress score of 20 — alongside a new steps-vs-engagement metric (4.86K). As before, when perceived stress score is 2 or lower, average engagement increases by 2.72 points, compared to 1.6 points when stress is between 2 and 7, reinforcing that lower-stress users tend to be more deeply engaged with the platform.

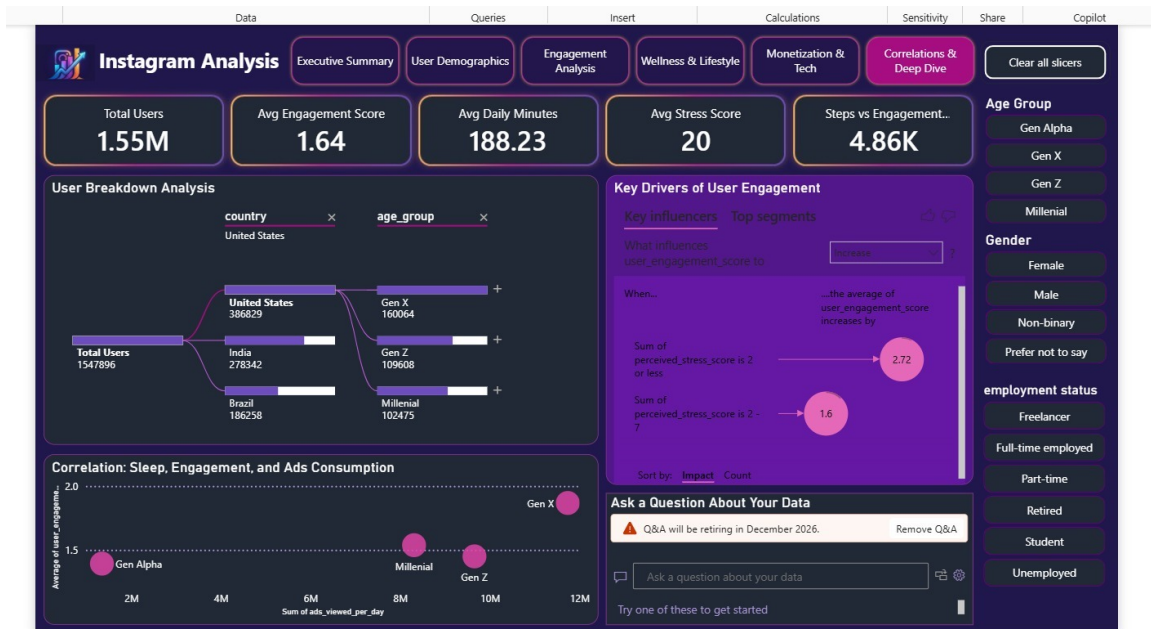


Figure 6 — Correlations & Deep Dive page

4. Key Performance Indicators (KPIs)

4.1 Data Cleaning & Processing — Tools: Python, Excel

- 100% of missing values handled.
- 100% of duplicate records removed.
- Dataset validated and converted to correct data types.

4.2 Analysis & Insights — Tools: Python, SQL

- ≥ 90% of analytical questions answered.
- At least 5 key insights identified.
- Relationships between social media usage and lifestyle/productivity indicators identified.

4.3 Visualization & Reporting — Tool: Power BI

- Dashboard load time under 3 seconds.
- At least 5 visualizations included per page (24+ across the report).

- $\geq 80\%$ usability for users navigating the dashboard.

4.4 Final Documentation & Presentation

- 100% completion of the final report.
- Clear explanation of the analysis process.
- At least 3 actionable recommendations delivered.

5. Roles & Responsibilities

Role	Team Members	Responsibilities
Data Cleaning & Python Analysis	Nour Ahmed Youssef, Mohamed Nagah Mohamed	Data preprocessing and exploratory analysis
Database & SQL Analysis	Abdelrahman Mohamed Hamza, Mahmoud Ahmed Abbas	SQL queries and data aggregation
Data Exploration	Mohamed Amr Mohamed, Youssef Emad Mohamady	Dataset exploration using Excel
Visualization	Entire Team	Power BI dashboard development

6. Recommendations

Based on the patterns observed across the dashboard — particularly the link between lower perceived stress and higher engagement, and the high average daily time spent in-app (188 minutes) — the team proposes the following actionable recommendations:

- **Limiting excessive social media usage** — encourage in-app screen-time reminders for users exceeding average daily minutes, particularly within the Gen Z and Millennial cohorts who show the highest message and like volumes.
- **Promoting digital wellbeing awareness** — surface sleep and stress-related nudges, since lower stress scores correlate with materially higher engagement, suggesting wellbeing and healthy platform use can reinforce each other rather than trade off.
- **Encouraging better time management** — highlight time-on-feed versus time-on-explore breakdowns to users so they can recognize passive scrolling patterns versus intentional content discovery.

7. Final Presentation

The team will present the full Power BI dashboard and analysis results described in this report, walking through the six report pages and highlighting the key insights and recommendations on how social media usage relates to lifestyle, wellbeing, and digital productivity.

8. SQL Validation Queries

To validate the aggregated metrics displayed on the Power BI dashboard, five T-SQL queries were run directly against the star-schema model in SQL Server Management Studio, joining fact_usage to the relevant dimension tables. Each query below cross-checks a specific finding referenced elsewhere in this report against the raw grouped aggregates.

8.1 Average Explore Time by Country

Joins fact_usage to dim_country, grouping by country and keeping only countries where average daily Explore time exceeds 10 minutes.

All ten countries clear the 10-minute threshold and converge tightly at roughly 37 minutes of average daily Explore time regardless of location. The United States (386,829 users), India (278,342), and Brazil (186,258) contribute the largest user counts, matching the geographic distribution reported in the Executive Summary (Section 3.1).

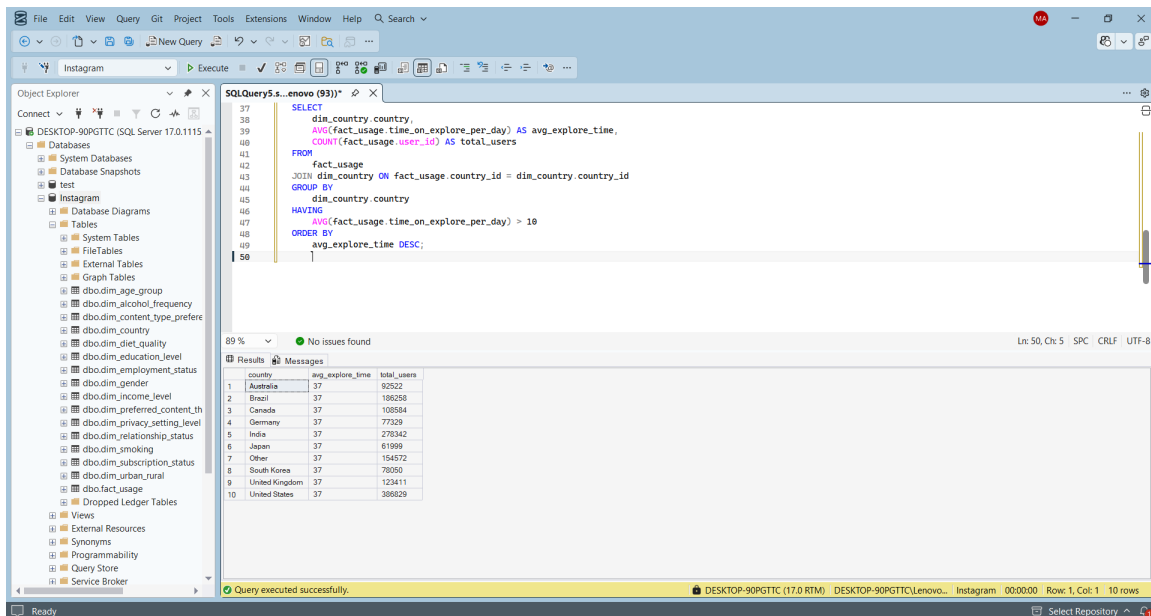


Figure 7 — Average Explore Time by Country (SQL validation)

8.2 User Engagement Score by Age Group

Joins `fact_usage` to `dim_age_group`, filtering to age groups with more than 100 users and ranking by average `user_engagement_score`.

Gen X posts the highest average engagement score (1.87), ahead of Millennials (1.54), Gen Z (1.45), and Gen Alpha (1.39). Because Gen X is also the largest cohort in the dataset (642K users, per Section 3.2), it likewise accounts for the greatest total volume of daily likes given (69.4M), consistent with the aggregate view shown on the updated Engagement Analysis page (Section 3.3).

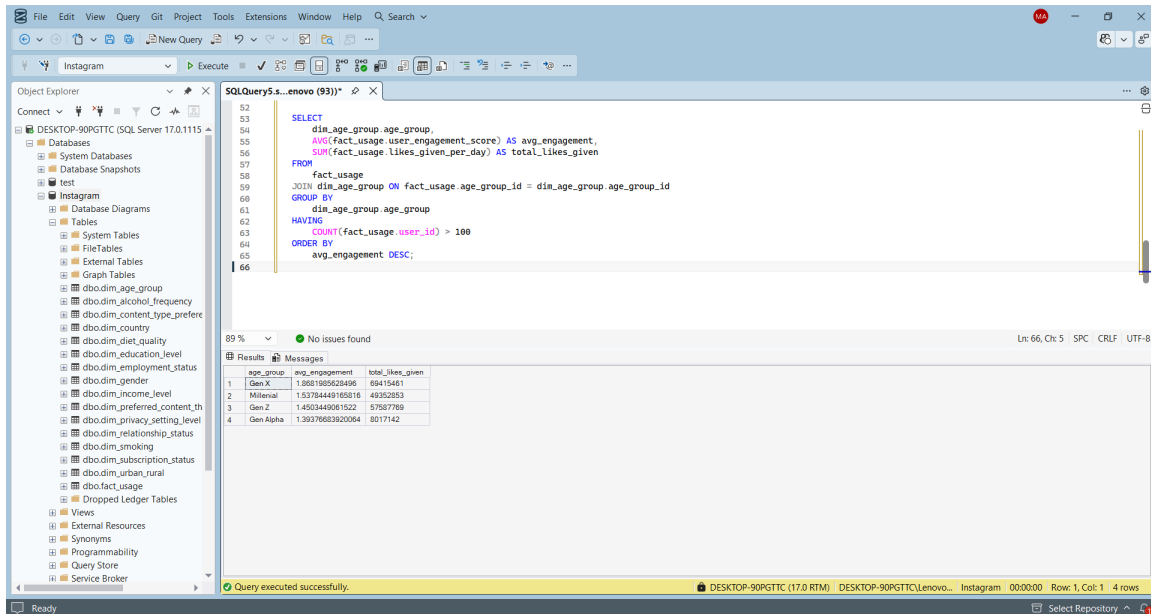


Figure 8 — User Engagement Score by Age Group (SQL validation)

8.3 Ad Interaction by Content Theme

Joins `fact_usage` to `dim_preferred_content_theme`, grouping by theme and keeping only themes with more than 500 total ads viewed.

Fitness (3.90M total ads viewed) and Fashion (3.89M) top the eight content themes, corroborating Section 3.3's identification of Fitness and Tech as the platform's most popular themes. Average ads clicked per day holds flat at 4 across every theme, suggesting theme preference shapes how much ad inventory a user is served rather than how often they click on it.

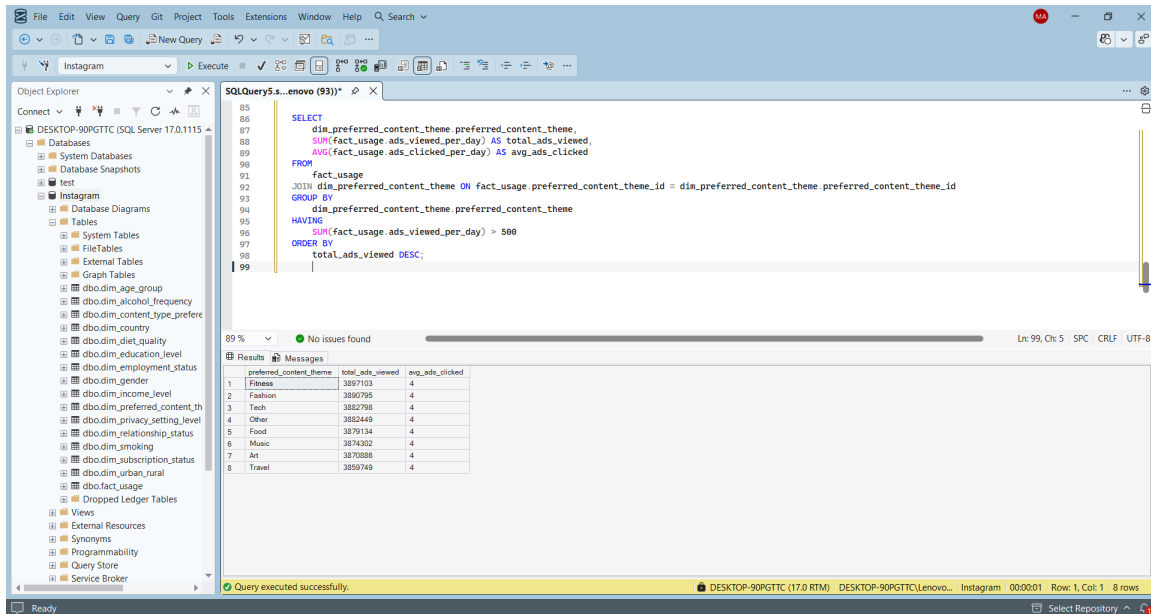


Figure 9 — Ad Interaction by Content Theme (SQL validation)

8.4 Premium Feature Usage by Income Level

Joins fact_usage to dim_income_level, restricting to users where uses_premium_features = 1 and ranking by average daily active minutes.

Among users with premium features enabled, Lower-middle income users log the highest average daily active minutes (188.4), narrowly ahead of Upper-middle (188.2) and Middle (188.2), while High-income premium users average the fewest (188.1). Middle-income accounts make up the largest premium segment by volume (102,481 users), consistent with Middle income being the leading income tier reported in Section 3.2.

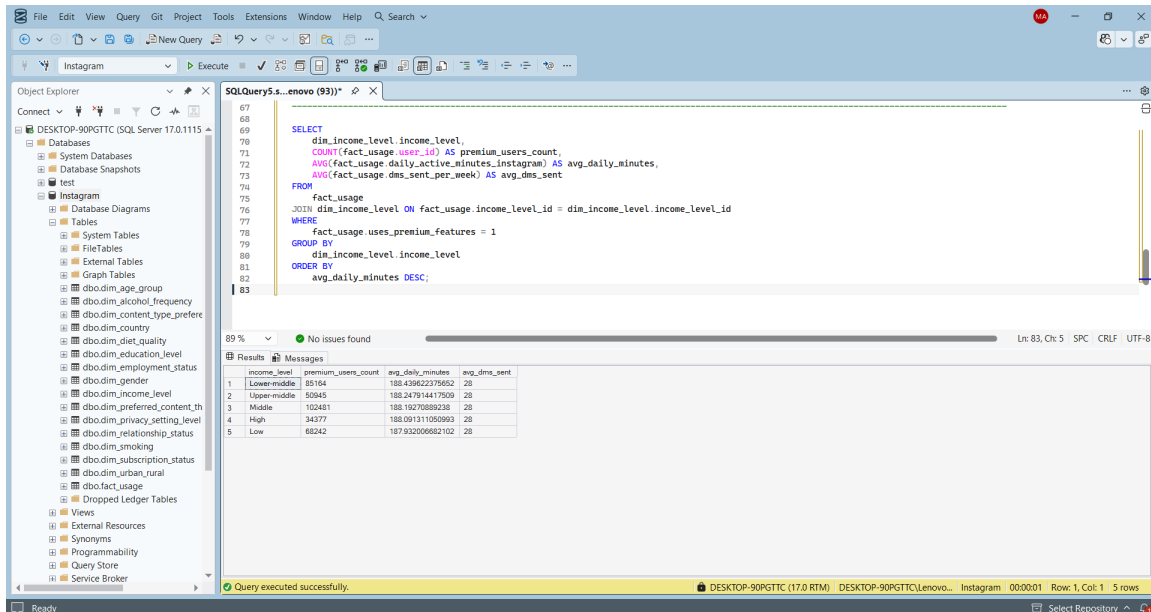


Figure 10 — Premium Feature Usage by Income Level (SQL validation)

8.5 Weekly Direct Messages by Relationship Status & Location

Joins `fact_usage` to both `dim_relationship_status` and `dim_urban_rural`, grouping by the combined pair and ranking by average weekly DMs sent.

Widowed users in Rural areas send the most direct messages per week on average (29), a point above every other relationship-status/location pairing, which converge tightly around 28 weekly DMs. Single users in Urban areas form the largest individual segment in the table (340,173 users), followed by Married and Divorced urban users.

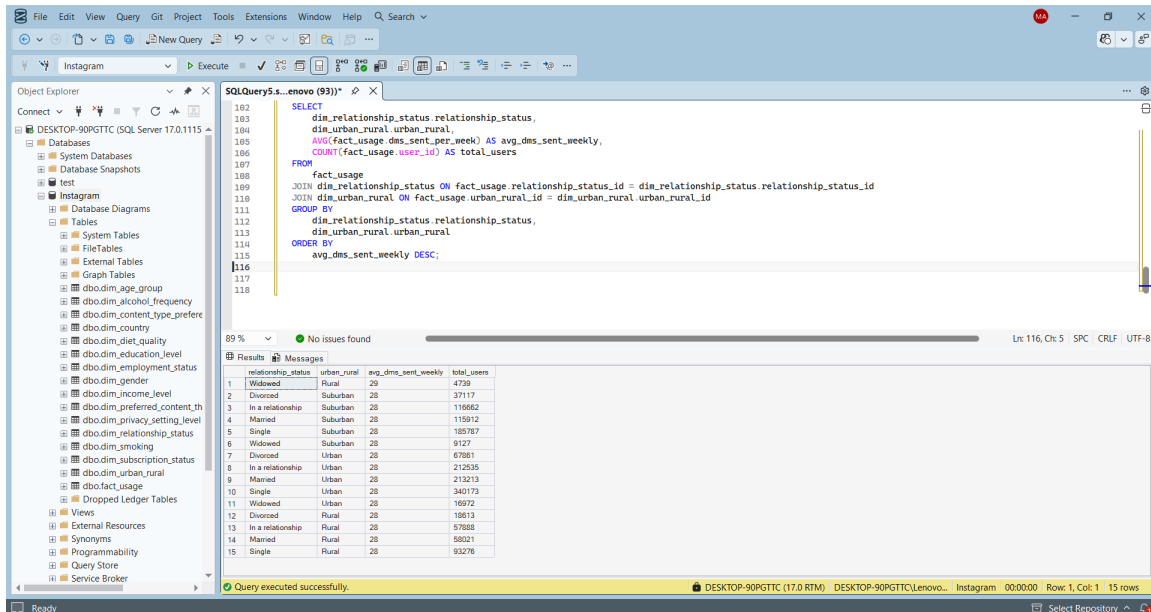


Figure 11 — Weekly Direct Messages by Relationship Status & Location (SQL validation)